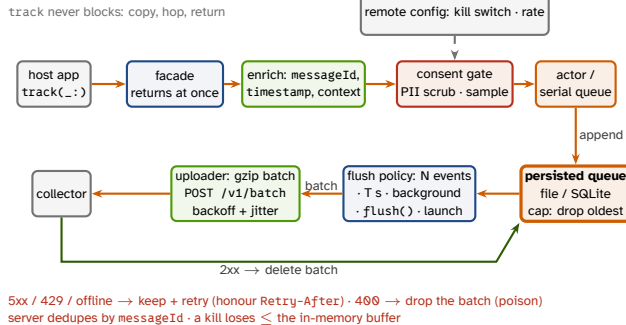


**In one line:** Same steps, two very different centres of gravity: **scope** → **functional** → **non-functional** → **data model** → **API** → **high-level** → **the hard parts**. An **SDK** lives inside *someone else's* app: it must never crash, block, bloat or leak it — the design is a **persisted, batched, privacy-gated queue**. An **E2EE messenger's** server is an untrusted **mailbox**: keys live only on devices, every message is sealed *per recipient device*, and the push extension has to decrypt.

### (a) Analytics SDK — the pipeline



### (a) Analytics SDK — step by step

**Scope** events + screens + identify; *not* dashboards, not crash reporting. Who integrates: other teams' apps, iOS 15+.

**Functional** track, screen, identify(userId:traits:), reset() on logout, flush(), opt-out.

**Non-functional** never crash / block the host; ≈0 launch cost; small binary; no data loss across kill/offline; bounded disk + battery; privacy by default; thread-safe.

**Data model** Event{messageId: UUID, name, properties, timestamp, context{app, os, device, locale, sessionId}, schemaVersion}; anonymousId until identify. sentAt added at upload → server corrects device-clock skew: receivedAt - (sentAt - timestamp).

**API** one entry point, value types, no callbacks required:

```
public final class Analytics: Sendable {
    public static func configure(writeKey: String, options: Options)
    public func track(_ name: String, _ props: [String: Value] = [:])
    public func identify(userId: String, traits: [String: Value])
    public func setConsent(_ c: Consent) // gates collection
    public func flush() // fire-and-forget
    public func reset() // logout: new anonymousId
}
```

### Deep dives

**Threading:** public calls copy the event and hop to one actor / serial queue — never main, never a lock the caller waits on. **Persistence:** append each event (or small batches) to a file / SQLite so a kill loses ≤ one in-memory batch; cap bytes + age, drop oldest. **Flush:** whichever first — N events (e.g. 20), T seconds (e.g. 30), app to background (inside beginBackgroundTask), explicit flush(), next launch. One upload in flight. **Retries:** exponential backoff + full jitter, capped; 5xx / 429 / offline = retry; a 4xx batch is *poison* — drop it or it blocks the queue forever. Idempotent: messageId dedup server-side. **Sampling:** decide per user/session with hash(id) % 100 < rate, not per event (keeps funnels whole); send the rate so the server re-weights. **Privacy:** no collection before consent; scrub PII (allow-list property keys, redact emails / phones); never IDFA without ATT; ship PrivacyInfo.xcprivacy (data types + required-reason APIs such as UserDefaults) and sign the binary. **Size + startup:** no heavy deps, nothing in +load / static initialisers, lazy I/O after launch; static XCFramework via SPM binary target. **Versioning:** SemVer; library evolution (BUILD\_LIBRARY\_FOR\_DISTRIBUTION); deprecate with @available(\*, deprecated, renamed:); schemaVersion in the payload so the server accepts old SDKs for years. **Kill switch:** remote config can stop sending. **Testing:** inject clock, Storage, Transport protocols; fake clock drives time-based flush; tests for kill mid-write, offline, 400 vs 500, consent off.

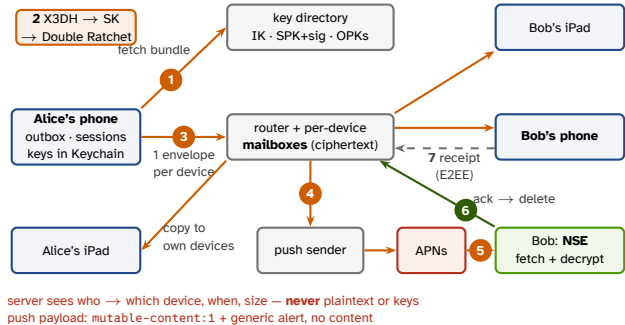
### Traps

- ! SDK: a fatalError / force-unwrap, sync disk I/O on the caller's thread, or work at launch — you just degraded every host app.
- ! SDK: retrying a 400 forever; in-memory-only queue (kill = data loss); per-event sampling.
- ! E2EE: keys in iCloud backup, or "the server encrypts it" (TLS ≠ E2EE).
- ! Curve25519 identity key "in the Secure Enclave" — SE keys are P-256 only; Keychain.
- ! Encrypting once per user, not per device — the iPad can't read it.
- ! Plaintext in the APNs payload — Apple and logs see it; the NSE decrypts.

### Trade-offs

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| queue    | append-only file: tiny, no deps · SQLite: queries, atomic batch delete          |
| binary   | static: no dyld cost at launch · dynamic: one copy shared with extensions       |
| upload   | URLSession + bg task: prompt · background session: survives a kill, delayed     |
| groups   | pairwise: N encryptions per send · sender keys: one, rekey when a member leaves |
| receipts | per message: simple · batched "read up to seq": far fewer messages              |

### (b) E2EE messenger — client ↔ server ↔ push



### (b) E2EE messenger — step by step

**Scope** 1:1 + small groups, text + media, multi-device; not calls (own design), not backups.

**Functional** send / receive offline, delivery + read receipts, history on each device, push.

**Non-functional** server can't read content; forward secrecy; ordering; works offline; low battery; notification shows plaintext.

**Data model** Message{clientId, conversationId, senderDevice, serverSeq?, body, state: sending|sent|delivered|read|failed}; Session per (user, device); Outbox row per send. Envelope on the wire: {to: userId, deviceId, type, ciphertext} — the server routes blobs. **API** PUT /keys (upload IK, signed prekey, 100 one-time prekeys) · GET /keys/{user} → bundle per device · POST /messages (envelopes for all recipient + own devices; 409 = device list stale) · WebSocket deliver/ack · GET /messages?since= catch-up.

### Deep dives

**Keys (Signal, conceptually):** each device has a long-term **identity key** (IK), a **signed prekey** (SPK, signed by IK, rotated) and one-time prekeys (OPK), published so a peer can start a session while it is *offline*. **X3DH:** Alice combines DH(IK<sub>A</sub>, SPK<sub>B</sub>), DH(EK<sub>A</sub>, IK<sub>B</sub>), DH(EK<sub>A</sub>, SPK<sub>B</sub>) [+ DH(EK<sub>A</sub>, OPK<sub>B</sub>)] through a KDF → shared secret SK; her first message carries IK<sub>A</sub>, EK<sub>A</sub> and which prekeys she used. (Signal now also specifies **PQXDH**, adding a post-quantum KEM.) **Double Ratchet:** SK seeds a *root chain*; a **symmetric ratchet** (KDF chain) gives every message its own key — delete after use = **forward secrecy**; a **DH ratchet** step each time a new ratchet public key arrives = **break-in recovery**. Header N / PN + stored skipped keys decrypt out-of-order messages. **Trust:** compare *safety numbers* (IK fingerprints, QR); warn on key change. **Multi-device** (Signal's Sesame): a session per device; sender encrypts once per recipient device and per own other device; stale devices kept briefly for late messages. Groups: pairwise fan-out, or *sender keys* for big groups. **Ordering + sync:** server assigns serverSeq per conversation on accept; client sorts by it, pending rows last; per-device mailbox deleted on ack; since catch-up on reconnect. **Receipts:** sent as encrypted control messages, batched; "read" is optional (privacy setting). **Push:** payload carries no content — mutable-content:1 + generic alert; the **NSE** fetches, decrypts with keys from a shared Keychain group (AfterFirstUnlockThisDeviceOnly) + App Group DB, rewrites the alert; timeout shows the generic text. NSE and app both advance the ratchet → cross-process lock, or the session corrupts. **Attachments:** random key per file, encrypt + MAC, upload ciphertext to a CDN, send {url, key, digest} *inside* the E2EE message.

### Remember

**SDK = quest: fast, silent, persisted, consented. E2EE = mailbox server: keys on devices, a session per device, the NSE decrypts.**